

IELTS

Workbook



Reading

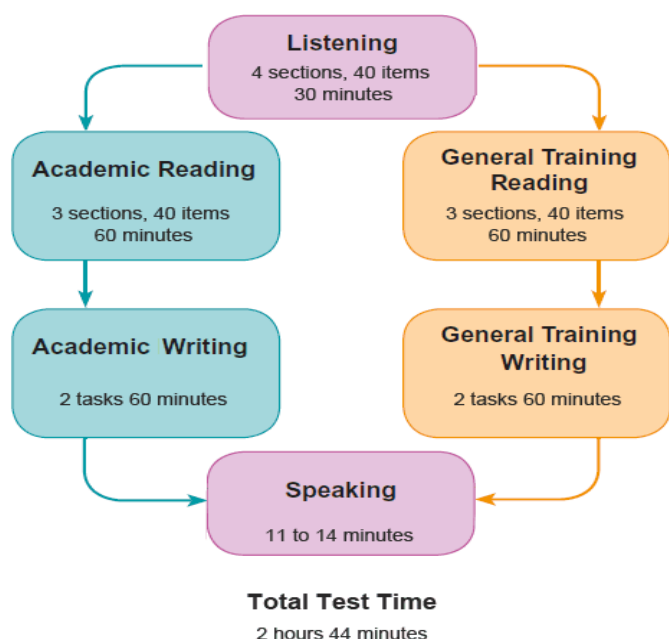
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Preface

IELTS (International English Language Testing System) is a worldwiderecognized official English exam. The IELTS is administered by the British Council, IDP: IELTS Australia and Cambridge English Language Assessment. Its aim is to evaluate the English level of the people that take it. This is one of the most accepted English exams by higher education institutions to certify English language proficiency.



IELTS Reading: Academic Module – 3 long texts of an academic nature. General Training Module - Mixture of long and short texts of a more general nature, as well as texts related to work situation.

Reading Paper is Assessed on:

- Identifying the overall purpose
- Follow key arguments in a text
- Identify opinions and attitudes
- Locate specific information
- Distinguish main ideas from details.
- Extract information from a text
- complete a diagram, summary, table

Reading Types of Questions:

- Matching Heading Questions
- True/ False/Not Given or Yes / No/ Not Given.
- Matching Paragraph Information Question.
- Summary Completion Question
- Sentence Completion Question
- Multiple Choice Question
- List Selection
- Choosing a Title
- Categorization Question
- Matching Sentence Endings
- Table Completion
- **Flow chart / Diagram Completion Question**
- Short Answer Question

A flow-chart / diagram is used to show the sequence of events in a process. Such questions are rarely asked but can account for 5-8 marks for a reading passage. Passages involving description of a process or a series of events mostly include such questions. Here the information may not be given in the same sequence as in the passage. Make sure you read the whole passage so that you can locate any keywords or paraphrases from the questions. With our tips & practise papers we are sure you're going to enjoy solving such question and get correct answers

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TO SOLVE FLOW-CHART QUESTIONS:

A flow-chart diagram is used to show the sequence of events in a process. Such questions are rarely asked but can account for 5-8 marks for a reading passage. Passages involving description of a process or a series of events mostly include such questions. Here the information may not be given in the same sequence as in the passage. Make sure you read the whole passage so that you can locate any keywords or paraphrases from the questions.

GUIDELINES TO SOLVE:FLOW-CHART COMPLETION QUESTIONS:

1. Skim read the entire passage to get the overall idea of the passage. (5 min)
2. Understand what kind of information is given in each paragraph. You can give headings to each paragraph for your self-understanding. This way it's easy for you to locate the paragraph which contains the information asked.
3. Study the flow-chart completion task and decide what type of information you need to find out.
4. Underline the relevant keywords of the statement to help you locate your answer.
5. Look for the paragraphs which have this text and also for keywords and paraphrases from the question. The cohesive devices like in addition to, firstly, secondly, furthermore, therefore, viz will help you understand the order of events.
6. Carefully read the text, before and after the words you have found.
7. Fill in the gaps with suitable words considering the word limit mentioned in the question.
8. Spend less than 1 min to analyse each statement given in the flow-chart.

Let's look at the application of the above-mentioned guidelines for the following passage below.

Solved Reading Passage

BAKELITE – The Birth Of Modern Plastics

In 1907, Leo Hendrick Baekeland, a Belgian scientist working in New York, discovered and patented a revolutionary new synthetic material. His invention, which he named 'Bakelite', was of enormous technological importance, and effectively launched the modern plastics industry.

The term 'plastic' comes from the Greek *plassein*, meaning 'to mould'. Some plastics are derived from natural sources, some are semi-synthetic (the result of chemical action on a natural substance), and some are entirely synthetic, that is, chemically engineered from the constituents of coal or oil. Some are 'thermoplastic', which means that, like candlewax, they melt when heated and can then be reshaped. Others are 'thermosetting': like eggs, they cannot revert to their original viscous state, and their shape is thus fixed for ever., Bakelite had the distinction of being the first totally synthetic thermosetting plastic.

The history of today's plastics begins with the discovery of a series of semi-synthetic thermoplastic materials in the mid-nineteenth century. The impetus behind the development of these early plastics was generated by a number of factors - immense technological progress in the domain of chemistry, coupled with wider cultural changes, and the pragmatic need to find acceptable substitutes for dwindling supplies of 'luxury' materials such as tortoiseshell and ivory.

Baekeland's interest in plastics began in 1885 when, as a young chemistry student in Belgium, he embarked on research into phenolic resins, the group of sticky substances produced when phenol (carbolic acid) combines with an aldehyde (a volatile fluid similar to alcohol). He soon abandoned the subject, however, only returning to it some years later. By 1905 he was a wealthy New Yorker, having recently made his fortune with the invention of a new photographic paper. While Baekeland had been busily amassing dollars, some advances had been made in the development of plastics. The years 1899 and 1900 had seen the patenting of the first semi-synthetic thermosetting material that could be manufactured on an industrial scale. In purely scientific terms, Baekeland's major contribution to the field is not so much the actual discovery of the material to which he gave his name, but rather the method by which a reaction between phenol and formaldehyde could be controlled, thus making possible its preparation on a commercial basis. On 13 July 1907, Baekeland took out his famous patent describing this preparation, the essential features of which are still in use today.

The original patent outlined a three-stage process, in which phenol and formaldehyde (from wood or coal) were initially combined under vacuum inside a large egg-shaped kettle. The result was a resin known as Novalak, which became soluble and malleable when heated. The resin was allowed to cool in shallow trays until it hardened, and then broken up and ground into powder. Other substances were then introduced: including fillers, such as woodflour, asbestos or cotton, which increase strength and moisture resistance, catalysts (substances to speed up the reaction between two chemicals without joining to either) and hexa, a compound of ammonia and formaldehyde which supplied the additional formaldehyde necessary to form a thermosetting

resin. This resin was then left to cool and harden, and ground up a second time. The resulting granular powder was raw Bakelite, ready to be made into a vast range of manufactured objects. In the last stage, the heated Bakelite was poured into a hollow mould of the required shape and subjected to extreme heat and pressure; thereby 'setting' its form for life.

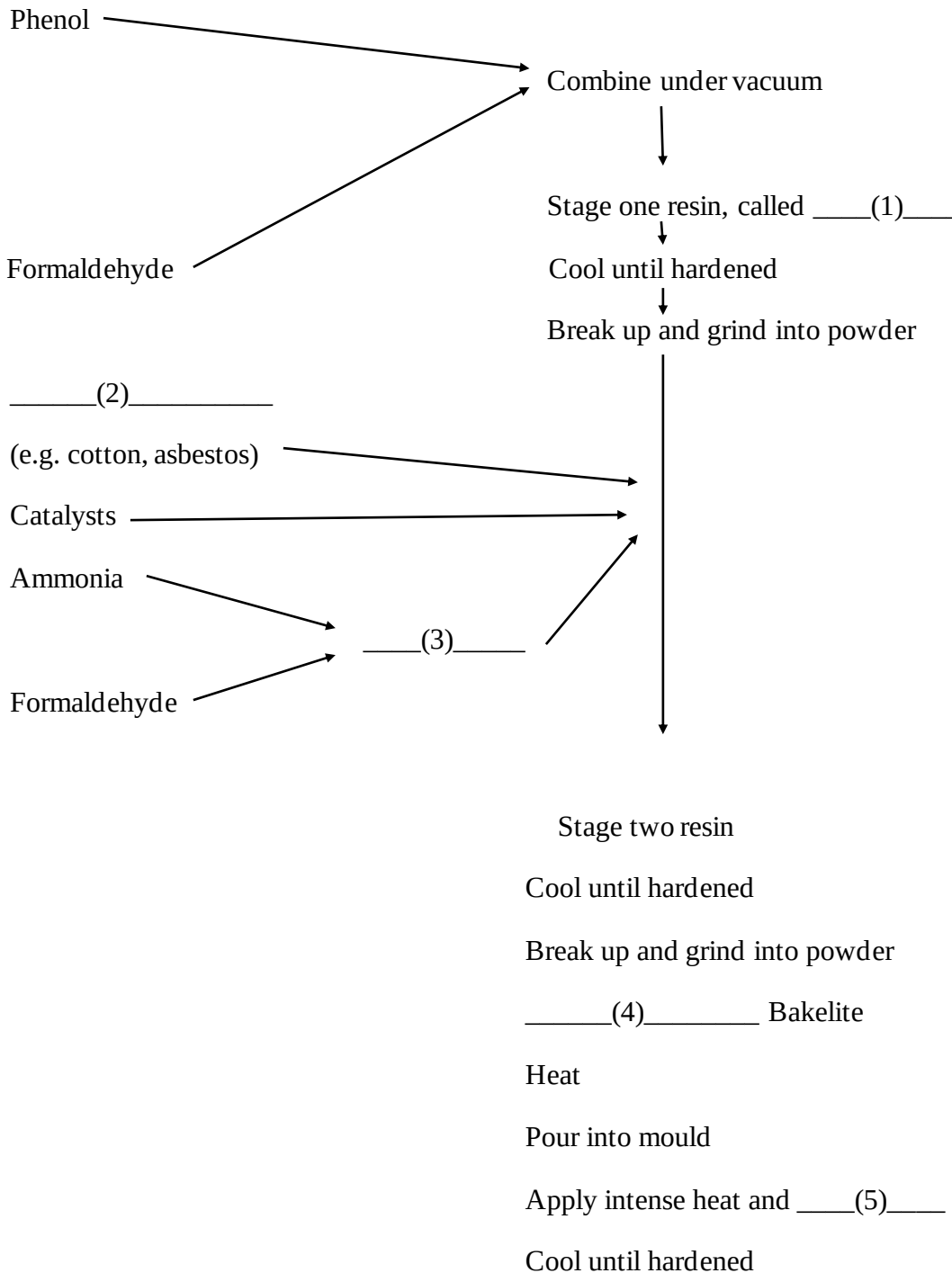
The design of Bakelite objects, everything from earrings to television sets, was governed to a large extent by the technical requirements of the moulding process. The object could not be designed so that it was locked into the mould and therefore difficult to extract. A common general rule was that objects should taper towards the deepest part of the mould, and if necessary the product was moulded in separate pieces. Moulds had to be carefully designed so that the molten Bakelite would flow evenly and completely into the mould. Sharp corners proved impractical and were thus avoided, giving rise to the smooth, 'streamlined' style popular in the 1930s. The thickness of the walls of the mould was also crucial: thick walls took longer to cool and harden, a factor which had to be considered by the designer in order to make the most efficient use of machines.

Baekeland's invention, although treated with disdain in its early years, went on to enjoy an unparalleled popularity which lasted throughout the first half of the twentieth century. It became the wonder product of the new world of industrial expansion - 'the material of a thousand uses'. Being both non-porous and heat-resistant, Bakelite kitchen goods were promoted as being germ-free and sterilisable. Electrical manufacturers seized on its insulating properties, and consumers everywhere relished its dazzling array of shades, delighted that they were now, at last, no longer restricted to the wood tones and drab browns of the prepfastic era. It then fell from favour again during the 1950s, and was despised and destroyed in vast quantities. Recently, however, it has been experiencing something of a renaissance, with renewed demand for original Bakelite objects in the collectors' marketplace, and museums, societies and dedicated individuals once again appreciating the style and originality of this innovative material.

Complete the flow-chart.

Choose ONE WORD ONLY from the passage for each answer. **The TWO questions below are solved for you.**

THE PRODUCTION OF BAKELITE



Solved examples:

1. This text belongs to the **paragraph E** which explains the three-stage process of Bakelite production. The first gap requires the stage one resin which is produced by combining phenol and formaldehyde under vacuum. The answer can be traced from the following text, ‘.....were initially combined under vacuum inside a large egg-shaped kettle. The result was a resin known as **Novalak**, which became soluble and malleable when heated. This indicates that the word for the fill in the gap is **Novalak**.
2. The second gap text shows the reference of fillers like asbestos, cotton or wood flour which are required for the second stage resin. The answer can be traced from the following text, ‘other substances were then introduced: including **fillers**, such as wood flour, asbestos or cotton, which increase strength and moisture resistance, necessary to form a thermosetting resin.’ This indicates that the word for the fill in the gap is **fillers**.

Practise Reading Passage 1.1

FLAWED BEAUTY: THE PROBLEM WITH TOUGHENED GLASS

On 2nd August 1999, a particularly hot day in the town of Cirencester in the UK, a large pane of toughened glass in the roof of a shopping centre at Bishops Walk shattered without warning and fell from its frame. When fragments were analysed by experts at the giant glass manufacturer Pilkington, which had made the pane, they found that minute crystals of nickel sulphide trapped inside the glass had almost certainly caused the failure.

‘The glass industry is aware of the issue,’ says Brian Waldron, chairman of the standards committee at the Glass and Glazing Federation, a British trade association, and standards development officer at Pilkington. But he insists that cases are few and far between. ‘It’s a very rare phenomenon,’ he says.

Others disagree. ‘On average I see about one or two buildings a month suffering from nickel sulphide related failures,’ says Barrie Josie, a consultant engineer involved in the Bishops Walk investigation. Other experts tell of similar experiences. Tony Wilmott of London-based consulting engineers Sandberg, and Simon Armstrong at CladTech Associates in Hampshire both say they know of hundreds of cases. ‘What you hear is only the tip of the iceberg,’ says Trevor Ford, a glass expert at Resolve Engineering in Brisbane, Queensland. He believes the reason is simple: ‘No-one wants bad press.’

Toughened glass is found everywhere, from cars and bus shelters to the windows, walls and roofs of thousands of buildings around the world. It’s easy to see why. This glass has five times the strength of standard glass, and when it does break it shatters into tiny cubes rather than large, razor-sharp shards. Architects love it because large panels can be bolted together to make transparent walls, and turning it into ceilings and floors is almost as easy.

It is made by heating a sheet of ordinary glass to about 620°C to soften it slightly, allowing its structure to expand, and then cooling it rapidly with jets of cold air. This causes the outer layer of the pane to contract and solidify before the interior. When the interior finally solidifies and shrinks, it exerts a pull on the outer layer that leaves it in permanent compression and produces a tensile force inside the glass. As cracks propagate best in materials under tension, the compressive force on the surface must be overcome before the pane will break, making it more resistant to cracking.

The problem starts when glass contains nickel sulphide impurities. Trace amounts of nickel and sulphur are usually present in the raw materials used to make glass, and nickel can also be introduced by fragments of nickel alloys falling into the molten glass. As the glass is heated, these atoms react to form tiny crystals of nickel sulphide. Just a tenth of a gram of nickel in the furnace can create up to 50,000 crystals

These crystals can exist in two forms: a dense form called the alpha phase, which is stable at high temperatures, and a less dense form called the beta phase, which is stable at room temperatures. The high temperatures used in the toughening process convert all the crystals to the dense, compact alpha form. But the subsequent cooling is so rapid that the crystals don't have time to change back to the beta phase. This leaves unstable alpha crystals in the glass, primed like a coiled spring, ready to revert to the beta phase without warning.

When this happens, the crystals expand by up to 4%. And if they are within the central, tensile region of the pane, the stresses this unleashes can shatter the whole sheet. The time that elapses before failure occurs is unpredictable. It could happen just months after manufacture, or decades later, although if the glass is heated – by sunlight, for example – the process is speeded up. Ironically, says Graham Dodd, of consulting engineers Arup in London, the oldest pane of toughened glass known to have failed due to nickel sulphide inclusions was in Pilkington's glass research building in Lathom, Lancashire. The pane was 27 years old.

Data showing the scale of the nickel sulphide problem is almost impossible to find. The picture is made more complicated by the fact that these crystals occur in batches. So even if on average, there is only one inclusion in 7 tonnes of glass, if you experience one nickel sulphide failure in your building, that probably means you've got a problem in more than one pane. Josie says that in the last decade he has worked on over 15 buildings with the number of failures into double figures.

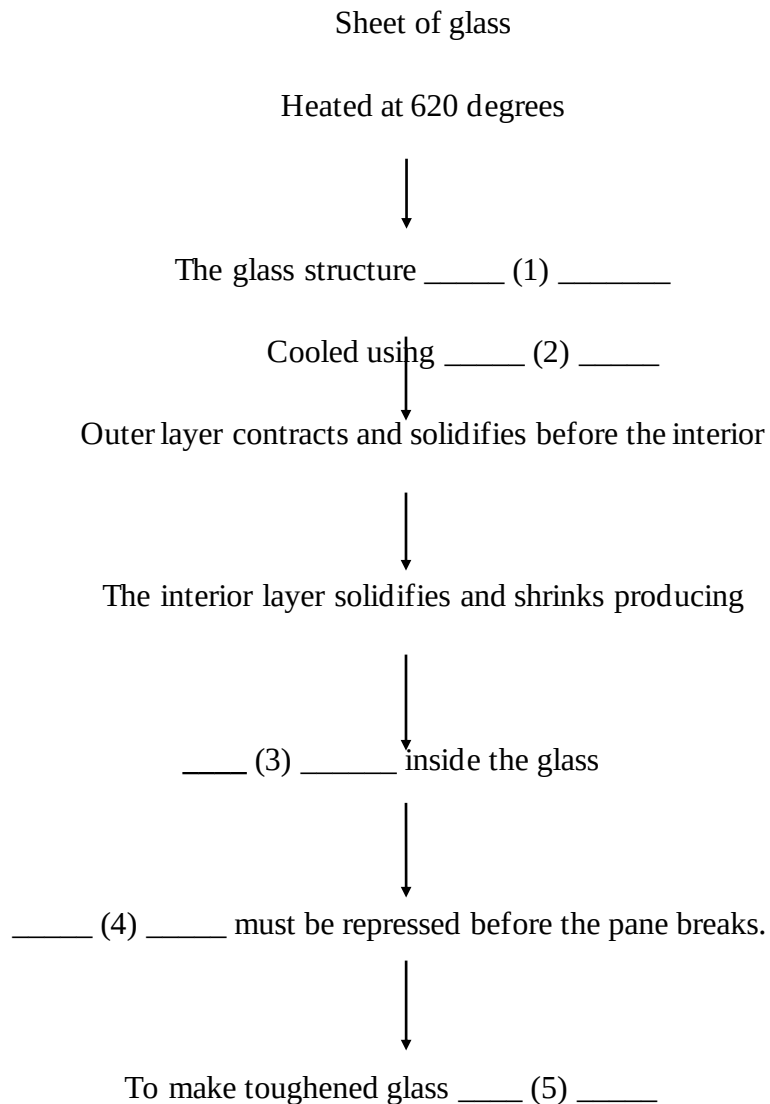
One of the worst examples of this is Waterfront Place, which was completed in 1990. Over the following decade the 40 storey Brisbane block suffered a rash of failures. Eighty panes of its toughened glass shattered due to inclusions before experts were finally called in. John Barry, an expert in nickel sulphide contamination at the University of Queensland, analysed every glass pane in the building. Using a studio camera, a photographer went up in a cradle to take photos of every pane

These were scanned under a modified microfiche reader for signs of nickel sulphide crystals. 'We discovered at least another 120 panes with potentially dangerous inclusions which were then replaced,' says Barry. 'It was a very expensive and time-consuming process that took around six months to complete.' Though the project cost A\$1.6 million (nearly £700,000), the alternative – re-cladding the entire building – would have cost ten times as much.

Questions 1-5

Complete the given flow-chart.

Choose NOT MORE THAN THREE WORDS ONLY from the passage for each answer.



1. _____
2. _____
3. _____
4. _____
5. _____

Practise Reading Passage 1.2

ROBOTS AT WORK

- A. The newspaper production process has come a long way from the old days when the paper was written, edited, typeset and ultimately printed in one building with the journalists working on the upper floors and the printing presses going on the ground floor. These days the editor, subeditors and journalists who put the paper together are likely to find themselves in a totally different building or maybe even in a different city. This is the situation which now prevails in Sydney. The daily paper is compiled at the editorial headquarters, known as the prepress centre, in the heart of the city, but printed far away in the suburbs at the printing centre. Here human beings are in the minority as much of the work is done by automated machines controlled by computers.
- B. Once the finished newspaper has been created for the next morning's edition, all the pages are transmitted electronically from the prepress centre to the printing centre. The system of transmission is an update on the sophisticated page facsimile system already in use in many other newspapers. An image setter at the printing centre delivers the pages as films. Each page takes less than a minute to produce, although for colour pages four versions, once each for black, cyan, magenta and yellow are sent. The pages are then processed into photographic negatives and the film is used to produce aluminium printing plates ready for the presses.
- C. A procession of automated vehicles is busy at the new printing centre where the Sydney Morning Herald is printed each day. With lights flashing and warning horns honking, the robots (to give them their correct name, the LGVs or laser guided vehicles) look for all the world like enthusiastic machines from a science fiction movie, as they follow their own random paths around the plant busily getting on with their jobs. Automation of this kind is now standard in all modern newspaper plants. The robots can detect unauthorised personnel and alert security staff immediately if they find an "intruder"; not surprisingly, tall tales are already being told about the machines starting to take on personalities of their own.
- D. The robots' principal job, however, is to shift the newsprint (the printing paper) that arrives at the plant in huge reels and emerges at the other end sometime later as newspapers. Once the size of the day's paper and the publishing order are determined at head office, the information is punched into the computer and the LGVs are programmed to go about their work. The LGVs collect the appropriate size paper reels and take them where they have to go. When the press needs another reel its computer alerts the LGV system. The Sydney LGVs move busily around the press room fulfilling their two key functions to collect reels of newsprint either from the reel stripping stations or from the racked supplies in the newsprint storage area. At the stripping station, the tough wrapping that helps to protect a reel of paper from rough handling is removed. Any damaged paper is peeled off and the reel is then weighed.

- E. Then one of the four paster-robots moves in. Specifically designed for the job, it trims the paper neatly and prepares the reel for the press. If required the reel can be loaded directly onto the press; if not needed immediately, an LGV takes it to the storage area. When the press computer calls for a reel, an LGV takes it to the reel loading area of the presses. It lifts the reel into the loading position and places it in the correct spot with complete accuracy. As each reel is used up, the press drops the heavy cardboard core into a waste bin. When the bin is full, another LGV collects it and deposits the cores into a shredder for recycling.
- F. The LGVs move at walking speed. Should anyone step in front of one or get too close, sensors stop the vehicle until the path is clear. The company has chosen a laser guide function system for the vehicles because, as the project development manager says “The beauty of it is that if you want to change the routes, you can work out a new route on your computer and lay it down for them to follow”. When an LGV’s batteries run low, it will take itself offline and go to the nearest battery maintenance point for replacement batteries. And all this is achieved with absolute minimum human input and a much-reduced risk of injury to people working in the printing centres.
- G. The question newspaper workers must now ask, however, is, “how long will it be before the robots are writing the newspapers as well as running the printing centre, churning out the latest edition every morning?”

Questions 1-8

Using the information in the passage, complete the flow chart below. Use **NO MORE THAN THREE WORDS** from the passage for each answer.

THE PRODUCTION PROCESS

The final version of the text is _____(1)_____ to the printing centre.



The pages converted into _____(2)_____.



_____(3)_____ are made use in the printing presses.



The LGV’s are _____(4)_____ by computer.



The LGV’s collect the reels of paper.



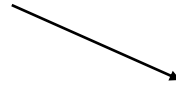
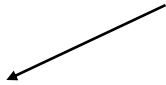
The LGV's remove the _____(5)_____ from the reel.



The reel is _____(6)_____



The reel is trimmed and prepared by the _____(7)_____.



The reel is taken to the press
_____(8)_____.

The reel is taken to the

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

Practise Reading Passage 1.3

THE HISTORY OF TORTOISE

If you go back far enough, everything lived in the sea. At various points in evolutionary history, enterprising individuals within many different animal groups moved out onto the land, sometimes even to the most parched deserts, taking their own private seawater with them in blood and cellular fluids. In addition to the reptiles, birds, mammals and insects which we see all around us, other groups that have succeeded out of water include scorpions, snails, crustaceans such as woodlice and land crabs, millipedes and centipedes, spiders and various worms. And we mustn't forget the plants, without whose prior invasion of the land none of the other migrations could have happened.

Moving from water to land involved a major redesign of every aspect of life, including breathing and reproduction. Nevertheless, a good number of thoroughgoing land animals later turned around, abandoned their hard-earned terrestrial re-tooling, and returned to the water. Seals have only gone part way back. They show us what the intermediates might have been like, on the way to extreme cases such as whales and dugongs. Whales (including the small whales we call dolphins) and dugongs, with their close cousins the manatees, ceased to be land creatures altogether and reverted to the full marine habits of their remote ancestors. They don't even come ashore to breed. They do, however, still breathe air, having never developed anything equivalent to the gills of their earlier marine incarnation. Turtles went back to the sea a very long time ago and, like all vertebrate returnees to the water, they breathe air. However, they are, in one respect, less fully given back to the water than whales or dugongs, for turtles still lay their eggs on beaches.

There is evidence that all modern turtles are descended from a terrestrial ancestor which lived before most of the dinosaurs. There are two key fossils called *Proganochelys quenstedti* and *Palaeochersis talampayensis* dating from early dinosaur times, which appear to be close to the ancestry of all modern turtles and tortoises. You might

Walter Joyce and Jacques Gauthier, at Yale University, obtained three measurements in these particular bones of 71 species of living turtles and tortoises. They used a kind of triangular graph paper to plot the three measurements against one another. All the land tortoise species formed a tight cluster of points in the upper part of the triangle; all the water turtles cluster in the lower part of the triangular graph. There was no overlap, except when they added some species that spend time both in water and on land. Sure enough, these amphibious species show up on the triangular graph approximately half way between the 'wet cluster' of sea turtles and the 'dry cluster' of land tortoises. The next step was to determine where the fossil fell. The bones of *P. quenstedti* and *P. talampayensis* leave us in no doubt. Their points on the graph are right in the thick of the dry cluster. Both these fossils were dry-land tortoises. They come from the era before our turtles returned to the water.

You might think, therefore, that modern land tortoises have probably stayed on land ever since those early terrestrial times, as most mammals did after a few of them went back to the sea. But apparently not. If you draw out the family tree of all modern turtles and tortoises, nearly all the branches are aquatic.

wonder how we can tell whether fossil animals lived on land or in water, especially if only fragments are found. Sometimes it's obvious. Ichthyosaurus were reptilian contemporaries of the dinosaurs, with fins and streamlined bodies. The fossils look like dolphins and they surely lived like dolphins, in the water. With turtles it is a little less obvious. One way to tell is by measuring the bones of their forelimbs.

Today's land tortoises constitute a single branch, deeply nested among branches consisting of aquatic turtles. This suggests that modern land tortoises have not stayed on land continuously since the time of *P. quenstedti* and *P. talampayensis*. Rather, their ancestors were among those who went back to the water, and they then re-emerged back onto the land in (relatively) more recent times.

Tortoises therefore represent a remarkable double return. In common with all mammals, reptiles and birds, their remote ancestors were marine fish and before that various more or less worm-like creatures stretching back, still in the sea, to the primeval bacteria. Later ancestors lived on land and stayed there for a very large number of generations. Later ancestors still evolved back into the water and became sea turtles. And finally, they returned yet again to the land as tortoises, some of which now live in the driest of deserts.

Questions 1-6

Complete the flow-chart below. Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer. Write your answers in boxes 1-6 on your answer sheet.

Method of determining where the ancestors of turtles and tortoises come from

Step 1: 71 species of living turtles and tortoises were examined and a total of **1**.....were taken from the bones of their forelimbs.



Step 2: The data was recorded on a **2** (necessary for comparing the information).
Outcome: Land tortoises were represented by a dense **3** of points towards the top. Sea turtles were grouped together in the bottom part.



Step 3: The same data was collected from some living **4** species and added to the other results. Outcome: The points for these species turned out to be positioned about **5** up the triangle between the land tortoises and the sea turtles.



Step 4: Bones of *R. quenstedti* and *P. talampayensis* were examined in a similar way and the results added.

Outcome: The position of the points indicated that both these ancient creatures were **6**.....

Practise Reading Passage 1.4

CAREERS WITH KIWI AIR Flight Attendants – Recruitment and Training Process

Recruitment

The position of Flight Attendant is one of prestige and immense responsibility. Recruitment is conducted according to operational demands and there can be periods of up to 12 months where no new intake is required. However, applications are always welcomed.

After you submit your initial application online, the Kiwi Air HR Services Team review the details you have provided. Candidates whose details closely match the requirements of the position are then contacted via email advising that their application has progressed to the next stage of the recruitment process. Potential candidates are then asked to attend a Walk-In Day. This could occur several weeks or months after the original application has been submitted depending on current needs.

The Walk-In Day consists of a brief presentation about the role and a short interview. Candidates who are successful on the Walk-In Day are notified within 10 days and invited to attend an Assessment Centre. Please note that candidates are required to pass a swimming test before attending the Assessment Centre. At the Assessment Centre, candidates attend an interview as well as participating in a number of assessments. Verbal references are then requested, and candidates attend a medical check.

At times, there may not be a need to recruit for Flight Attendant positions. However, the company continuously maintains a ‘recruitment pool’ of those who have completed the Assessment Centre stage. These candidates are contacted when a need for Flight Attendants is established, and attend a full interview before a decision is made on whether to extend an offer of employment.

Due to the volume of applications received, Kiwi Air is not able to offer verbal feedback to candidates at any stage of the recruitment process. Unsuccessful candidates may reapply at any time after 12 months from the date at which their applications are declined.

Training

Upon being offered a role as a trainee Flight Attendant, a 5-week training course is undertaken at our Inflight Services Training Centre in Auckland. This covers emergency procedures, customer care and service delivery, and equipment knowledge. To successfully complete the course, high standards must be attained and maintained in all subjects.

Questions 7-13

Complete the table below. Choose **NO MORE THAN THREE WORDS** from the text for each answer.

Write your answers in boxes **1-7** on your answer sheet.

Candidates go online to complete their **1**_____.



Suitable candidates are then invited to come to a **2**_____.



After having satisfactorily completed a **3**_____, successful candidates will then go to an Assessment Centre.



Kiwi Air then asks for **4**_____ and candidates are required to undergo a medical check.



If there is no immediate need for flight attendants, successful candidates are put into a **5**_____.



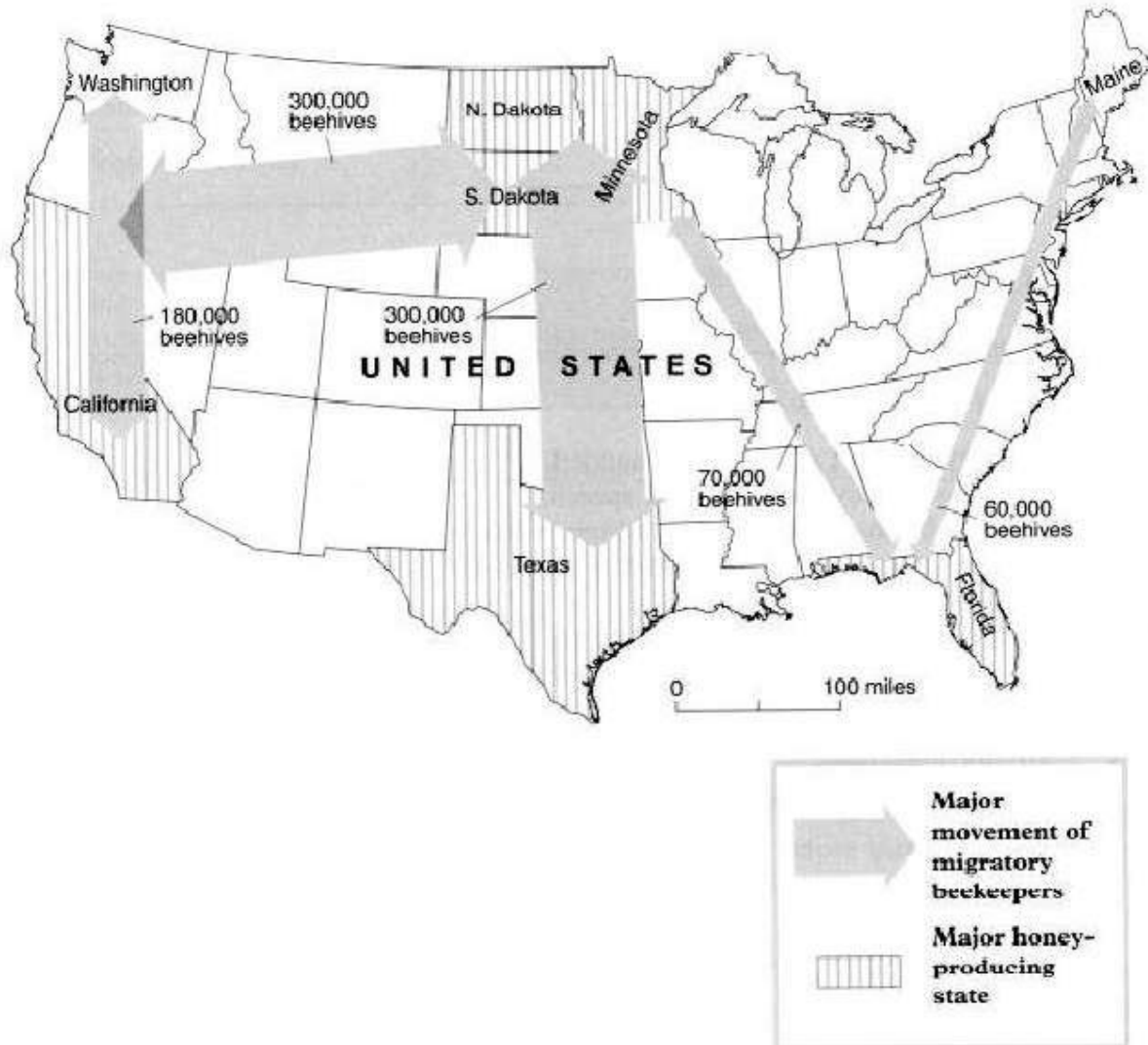
When the need arises, these candidates will then be given a **6**_____, after which they may be offered a job.



On starting the job, a 5-week training programme is given which includes how to look after passengers and what to do in an **7**_____.

Practise Reading Passage 1.5

MIGRATORY BEEKEEPING



Taking Wing

To eke out a full-time living from their honeybees, about half the nation's 2,000 commercial beekeepers pull up stakes each spring, migrating north to find more flowers for their bees. Besides turning floral nectar into honey, these hardworking insects also pollinate crops for farmers -for a fee. As autumn approaches, the beekeepers pack up their hives and go south, scrambling for pollination contracts in hot spots like California's fertile Central Valley.

Of the 2,000 commercial beekeepers in the United States about half migrate. This pays off in two ways. Moving north in the summer and south in the winter lets bees work a longer blooming season, making more honey — money — for their keepers. Second, beekeepers can carry their hives to farmers who need bees to pollinate their crops. Every spring a migratory beekeeper in California may move up to 160 million bees to flowering fields in Minnesota and every winter his

family may haul the hives back to California, where farmers will rent the bees to pollinate almond and cherry trees.

Migratory beekeeping is nothing new. The ancient Egyptians moved clay hives, probably on rafts, down the Nile to follow the bloom and nectar flow as it moved toward Cairo. In the 1880s North American beekeepers experimented with the same idea, moving bees on barges along the Mississippi and on waterways in Florida, but their lighter, wooden hives kept falling into the water. Other keepers tried the railroad and horse-drawn wagons, but that didn't prove practical. Not until the 1920s when cars and trucks became affordable and roads improved, did migratory beekeeping begin to catch on.

For the Californian beekeeper, the pollination season begins in February. At this time, the beehives are in particular demand by farmers who have almond groves; they need two hives an acre. For the three-week long bloom, beekeepers can hire out their hives for \$32 each. It's a bonanza for the bees too. Most people consider almond honey too bitter to eat so the bees get to keep it for themselves.

By early March it is time to move the bees. It can take up to seven nights to pack the 4,000 or so hives that a beekeeper may own. These are not moved in the middle of the day because too many of the bees would end up homeless. But at night, the hives are stacked onto wooden pallets, back-to-back in sets of four, and lifted onto a truck. It is not necessary to wear gloves or a beekeeper's veil because the hives are not being opened and the bees should remain relatively quiet. Just in case some are still lively, bees can be pacified with a few puffs of smoke blown into each hive's narrow entrance.

In their new location, the beekeeper will pay the farmer to allow his bees to feed in such places as orange groves. The honey produced here is fragrant and sweet and can be sold by the beekeepers. To encourage the bees to produce as much honey as possible during this period, the beekeepers open the hives and stack extra boxes called supers on top. These temporary hive extensions contain frames of empty comb for the bees to fill with honey. In the brood chamber below, the bees will stash honey to eat later. To prevent the queen from crawling up to the top and laying eggs, a screen can be inserted between the brood chamber and the supers. Three weeks later the honey can be gathered.

Foul smelling chemicals are often used to irritate the bees and drive them down into the hive's bottom boxes, leaving the honey-filled supers more or less bee free. These can then be pulled off the hive. They are heavy with honey and may weigh up to 90 pounds each. The supers are taken to a warehouse. In the extracting room, the frames are lifted out and lowered into an "uncapper" where rotating blades shave away the wax that covers each cell. The uncapped frames are put in a carousel that's on the bottom of a large stainless steel drum. The carousel is filled to capacity with 72 frames. A switch is flipped and the frames begin to whirl at 300 revolutions per minute;

centrifugal force throws the honey out of the combs. Finally, the honey is poured into barrels for shipment.

After this, approximately a quarter of the hives weakened by disease, mites, or an ageing or dead queen, will have to be replaced. To create new colonies, a healthy double hive, teeming with bees, can be separated into two boxes. One-half will hold the queen and a young, already mated queen can be put in the other half, to make two hives from one. By the time the flowers bloom, the new queens will be laying eggs, filling each hive with young worker bees. The beekeeper's family will then migrate with them to their summer location.

Questions 1-7

The flow chart below outlines the movements of the migratory beekeeper as described in Reading Passage

Complete the flow chart Choose your answers from the box at the bottom of the page and write your answers in boxes 1-7 on your answer sheet.

BEEKEEPER MOVEMENTS

In March, beekeepers..... (1)..... for migration at night when the hives are (2) and the bees are generally tranquil. A little..... (3) can ensure that this is the case.



They transport their hives to orange groves where farmers..... (4) beekeepers for placing them on their land. Here the bees make honey.



After three weeks, the supers can be taken to a warehouse where..... (5) are used to remove the wax and extract the honey from the..... (6)..... .



After the honey collection, the old hives are rejected. Good double hives are..... (7)..... and re-queened and the beekeeper transports them to their summer base.

List of Words/Phrases

smoke	chemicals	pay
barrels	protection	charge
set off	light	split
pollinate	machines	supers
combs	screen	prepare
full	empty	queens

TO SOLVE DIAGRAM COMPLETION QUESTIONS:

In diagram labelling task, you will see a diagram and description of a process. Such questions include details from a particular part of the passage which should be read carefully. Most of the time the description of the process consists of the functions of the labelled parts or their structure or their connectivity (in case of any mechanical instrument) and accordingly these questions are designed. Such questions are rarely asked and account for about 3-4 marks in a reading test. These are comparatively easier and can be solved first.

GUIDELINES TO SOLVE:

1. Skim read the entire passage to get the overall idea of the passage. (5 min).
2. Understand what kind of information is given in each paragraph. You can give headings to each paragraph for your self-understanding. This way it's easy for you to locate the paragraph which contains the information asked.
3. Look at the diagram. Study it well. Pay attention to what kind of details are required.
4. Read that part of the passage which describes the process carefully.
5. Usually The words required to fill in the blanks are names of the parts the diagram or a word that describes its function.
6. Fill in the gaps with suitable words considering the word limit mentioned in the question.
7. Carefully examine the diagram, before and after the words you have found.
8. Spend less than 5 min to solve the diagram completion task.

Let's look at the application of the above-mentioned guidelines for the following passage below.

Solved Reading Passage

WHAT'S SO FUNNY? - John McCrone reviews recent research on humour

The joke comes over the headphones: 'Which side of a dog has the most hair? The left.' No, not funny. Try again. 'Which side of a dog has the most hair? The outside.' Hah! The punchline is silly yet fitting, tempting a smile, even a laugh. Laughter has always struck people as deeply mysterious, perhaps pointless. The writer Arthur Koestler dubbed it the luxury reflex: 'unique in that it serves no apparent biological purpose'.

Theories about humour have an ancient pedigree. Plato expressed the idea that humour is simply a delighted feeling of superiority over others. Kant and Freud felt that joke-telling relies on building up a psychic tension which is safely punctured by the ludicrousness of the punchline. But most modern humour theorists have settled on some version of Aristotle's belief that jokes are based on a reaction to or resolution of incongruity, when the punchline is either a nonsense or, though appearing silly, has a clever second meaning.

Graeme Ritchie, a computational linguist in Edinburgh, studies the linguistic structure of jokes in order to understand not only humour but language understanding and reasoning in machines. He says that while there is no single format for jokes, many revolve around a sudden and surprising conceptual shift. A comedian will present a situation followed by an unexpected interpretation that is also apt.

So even if a punchline sounds silly, the listener can see there is a clever semantic fit and that sudden mental 'Aha!' is the buzz that makes us laugh. Viewed from this angle, humour is just a form of creative insight, a sudden leap to a new perspective.

However, there is another type of laughter, the laughter of social appeasement and it is important to understand this too. Play is a crucial part of development in most young mammals. Rats produce ultrasonic squeaks to prevent their scuffles turning nasty. Chimpanzees have a 'play-face' - a gaping expression accompanied by a panting 'ah, ah' noise. In humans, these signals have mutated into smiles and laughs. Researchers believe social situations, rather than cognitive events such as jokes, trigger these instinctual markers of play or appeasement. People laugh on fairground rides or when tickled to flag a play situation, whether they feel amused or not.

Both social and cognitive types of laughter tap into the same expressive machinery in our brains, the emotion and motor circuits that produce smiles and excited vocalisations. However, if cognitive laughter is the product of more general thought processes, it should result from more expansive brain activity.

Psychologist Vinod Goel investigated humour using the new technique of 'single event' functional magnetic resonance imaging (fMRI). An MRI scanner uses magnetic fields and radio waves to track the changes in oxygenated blood that accompany mental activity. Until recently, MRI scanners needed several minutes of activity and so could not be used to track

rapid thought processes such as comprehending a joke. New developments now allow half-second 'snapshots' of all sorts of reasoning and problem-solving activities.

Although Goel felt being inside a brain scanner was hardly the ideal place for appreciating a joke, he found evidence that understanding a joke involves a widespread mental shift. His scans showed that at the beginning of a joke the listener's prefrontal cortex lit up, particularly the right prefrontal believed to be critical for problem solving. But there was also activity in the temporal lobes at the side of the head (consistent with attempts to rouse stored knowledge) and in many other brain areas. Then when the punchline arrived, a new area sprang to life -the orbital prefrontal cortex. This patch of brain tucked behind the orbits of the eyes is associated with evaluating information.

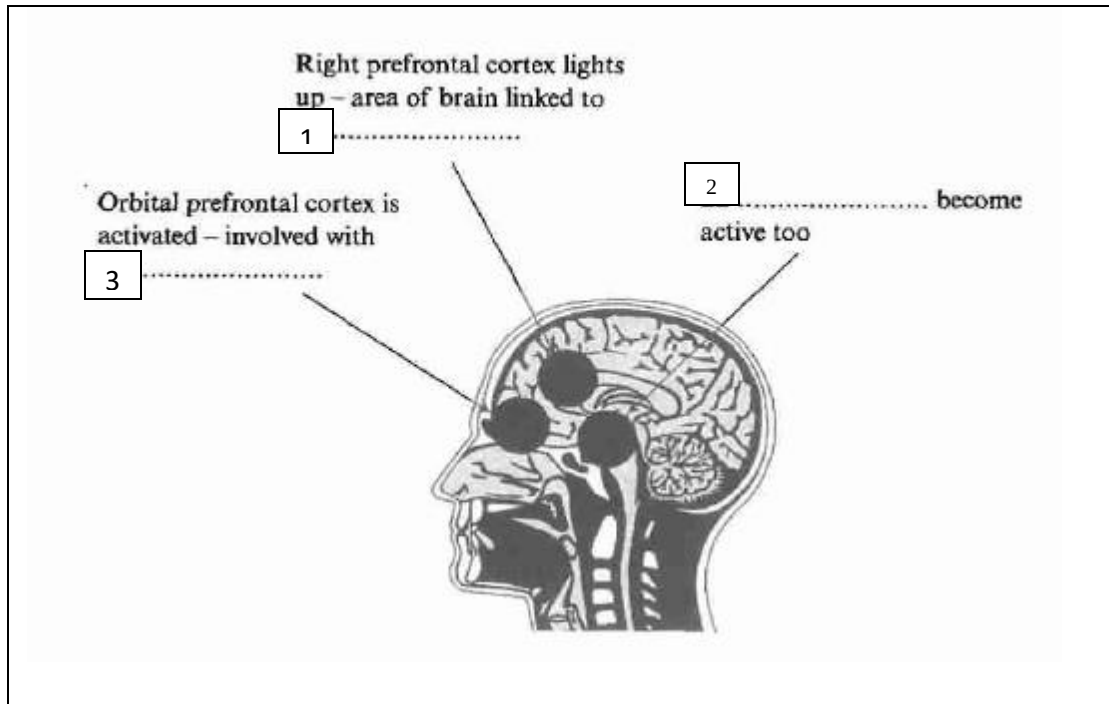
Making a rapid emotional assessment of the events of the moment is an extremely demanding job for the brain, animal or human. Energy and arousal levels may need, to be retuned in the blink of an eye. These abrupt changes will produce either positive or negative feelings. The orbital cortex, the region that becomes active in Goel's experiment, seems the best candidate for the site that feeds such feelings into higher-level thought processes, with its close connections to the brain's sub-cortical arousal apparatus and centres of metabolic control.

All warm-blooded animals make constant tiny adjustments in arousal in response to external events, but humans, who have developed a much more complicated internal life as a result of language, respond emotionally not only to their surroundings, but to their own thoughts. Whenever a sought-for answer snaps into place, there is a shudder of pleased recognition. Creative discovery being pleasurable, humans have learned to find ways of milking this natural response. The fact that jokes tap into our general evaluative machinery explains why the line between funny and disgusting, or funny and frightening, can be so fine. Whether a joke gives pleasure or pain depends on a person's outlook.

Humour may be a luxury, but the mechanism behind it is no evolutionary accident. As Peter Derks, a psychologist at William and Mary College in Virginia, says: 'I like to think of humour as the distorted mirror of the mind. It's creative, perceptual, analytical and lingual. If we can figure out how the mind processes humour, then we'll have a pretty good handle on how it works in general.'

The diagram below shows the areas of brain activated by jokes. **Label the diagram.**

Choose NO MORE THAN TWO WORDS from the passage for each answer. **One question below is solved for you.**



Solved Examples:

1. This text belongs to the **paragraph H** which mentions about the areas of brain activated by jokes. The answer for the first blank can be traced from the below given text from the passage which says, ‘His scans showed that at the beginning of a joke the listener’s prefrontal cortex lit up, particularly the right prefrontal believed to be **critical for problem solving**.’ This indicates that the word for the fill in the gap is **problem solving**.

Practise Reading Passage 1.1

VISUAL SYMBOLS AND THE BLIND

Part 1

From a number of recent studies, it has become clear that blind people can appreciate the use of outlines and perspectives to describe the arrangement of objects and other surfaces in space. But pictures are more than literal representations. This fact was drawn to my attention dramatically



Fig. 1

when a blind woman in one of my investigations decided on her own initiative to draw a wheel as it was spinning. To show this motion, she traced a curve inside the circle (Fig. 1). I was taken aback, lines of motion, such as the one she used, are a very recent invention in the history of illustration. Indeed, as art scholar David Kunzle notes, Wilhelm Busch, a trend-setting nineteenth-century cartoonist, used virtually no motion lines in his popular figure until about 1877.

When I asked several other blind study subjects to draw a spinning wheel, one particularly clever rendition appeared repeatedly: several subjects showed the wheel's spokes as curves lines. When asked about these curves, they all described them as metaphorical ways of suggesting motion. Majority rule would argue that this device somehow indicated motion very well. But was it a better indicator than, say, broken or wavy lines or any other kind of line, for that matter? The answer was not clear. So, I decided to test whether various lines of motion were apt ways of showing movement or if they were merely idiosyncratic marks. Moreover, I wanted to discover whether there were differences in how the blind and the sighted interpreted lines of motion.

To search out these answers, I created raised-line drawings of five different wheels, depicting spokes with lines that curved, bent, waved, dashed and extended beyond the perimeters of the wheel. I then asked eighteen blind volunteers to feel the wheels and assign one of the following motions to each wheel: wobbling, spinning fast, spinning steadily, jerking or braking. My control group consisted of eighteen sighted undergraduates from the University of Toronto.

All but one of the blind subjects assigned distinctive motions to each wheel. Most guessed that the curved spokes indicated that the wheel was spinning steadily; the wavy spokes, they thought; suggested that the wheel was wobbling, and the bent spokes were taken as a sign that the wheel was jerking. Subjects assumed that spokes extending beyond the wheel's perimeter signified that the wheel had its brakes on and that dashed spokes indicated the wheel was spinning quickly.

In addition, the favoured description for the sighted was favoured description for the blind in every instance. What is more, the consensus among the sighted was barely higher than that among the blind. Because motion devices are unfamiliar to the blind, the task I gave them involved some problem solving. Evidently, however, the blind not only figured out the meaning for each of the

motion, but as a group they generally came up with the same meaning at least as frequently as did sighted subjects.

Part2

We have found that the blind understand other kinds of visual metaphors as well. One blind woman drew a picture of a child inside a heart-choosing that symbol, she said, to show that love surrounded the child. With Chang Hong Liu, a doctoral student from china, I have begun exploring how well blind people understand the symbolism behind shapes such as hearts that do not directly represent their meaning.

We gave a list of twenty pairs of words to sighted subjects and asked them to pick from each pair the term that best related to a circle and the term that best related to square. For example, we asked: what goes with soft? A circle or a square? Which shape goes with hard?

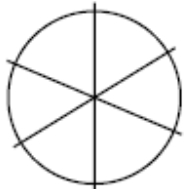
Words associated with circle/square	Agreement among subjects (%)
SOFT-HARD	100
MOTHER-FATHER	94
HAPPY-SAD	94
GOOD-EVIL	89
LOVE-HATE	89
ALIVE-DEAD	87
BRIGHT-DARK	87
LIGHT-HEAVY	85
WARM-COLD	81
SUMMER-WINTER	81
WEAK-STRONG	79
FAST-SLOW	79
CAT-DOG	74
SPRING-FALL	74
QUIET-LOUD	62
WALKING-STANDING	62
ODD-EVEN	57
FAR-NEAR	53
PLANT-ANIMAL	53
DEEP-SHALLOW	51

Fig. 2- Subjects were asked which word in each pair fits with a circle and which with a square. These percentages show the level of consensus among sighted subjects.

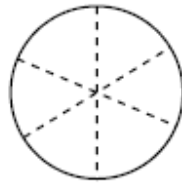
All our subjects deemed the circle soft and the square hard. A full 94% ascribed happy to the circle, instead of sad. But other pairs revealed less agreement: 79% matched fast to slow and weak to strong, respectively. And only 51% linked deep to circle and shallow to square. (See Fig. 2) When we tested four totally blind volunteers using the same list, we found that their choices closely resembled those made by the sighted subjects. One man, who had been blind since birth, scored extremely well. He made only one match differing from the consensus, assigning 'far' to square and 'near' to circle. In fact, only a small majority of sighted subjects, 53%, had paired far and near to the opposite partners. Thus we concluded that the blind interprets abstract shapes as sighted people do.

Questions 1-3

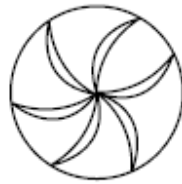
Look at the following diagrams (Questions 1-3), and the list of types of movement below. Match each diagram to the type of movement **A–E** generally assigned to it in the experiment. Choose the correct letter **A–E** and write them in boxes **1-3** on your answer sheet.



1



2



3

- A steady spinning
- B jerky movement
- C rapid spinning
- D wobbling movement
- E use of brakes

Practise Reading Passage 1.2

A REMARKABLE BEETLE

Some of the most remarkable beetles are the dung beetles, which spend almost their whole lives eating and breeding in dung’.

More than 4,000 species of these remarkable creatures have evolved and adapted to the world’s different climates and the dung of its many animals. Australia’s native dung beetles are scrub and woodland dwellers, specialising in coarse marsupial droppings and avoiding the soft cattle dung in which bush flies and buffalo flies breed.

In the early 1960s George Bornemissza, then a scientist at the Australian Government’s premier research organisation, the Commonwealth Scientific and Industrial Research Organisation (CSIRO), suggested that dung beetles should be introduced to Australia to control dung-breeding flies. Between 1968 and 1982, the CSIRO imported insects from about 50 different species of dung beetle, from Asia, Europe and Africa, aiming to match them to different climatic zones in Australia. Of the 26 species that are known to have become successfully integrated into the local environment, only one, an African species released in northern Australia, has reached its natural boundary.

Introducing dung beetles into a pasture is a simple process: approximately 1,500 beetles are released; a handful at a time, into fresh cow pats in the cow pasture. The beetles immediately disappear beneath the pats digging and tunnelling and, if they successfully adapt to their new environment, soon become a permanent, self-sustaining part of the local ecology. In time they multiply and within three or four years the benefits to the pasture are obvious.

Dung beetles work from the inside of the pat so they are sheltered from predators such as birds and foxes. Most species burrow into the soil and bury dung in tunnels directly underneath the pats, which are hollowed out from within. Some large species originating from France excavate tunnels to a depth of approximately 30 cm below the dung pat. These beetles make sausage-shaped brood chambers along the tunnels. The shallowest tunnels belong to a much smaller Spanish species that buries dung in chambers that hang like fruit from the branches of a pear tree. South African beetles dig narrow tunnels of approximately 20 cm below the surface of the pat. Some surface-dwelling beetles, including a South African species, cut perfectly-shaped balls from the pat, which are rolled away and attached to the bases of plants.

For maximum dung burial in spring, summer and autumn, farmers require a variety of species with overlapping periods of activity. In the cooler environments of the state of Victoria, the large French species (2.5 cms long) is matched with smaller (half this size), temperate-climate Spanish species. The former are slow to recover from the winter cold and produce only one or two generations of offspring from late spring until autumn. The latter, which multiplies rapidly in early spring, produce two to five generations annually. The South African ball-rolling species, being a subtropical beetle, prefers the climate of northern and coastal New South Wales where it commonly works with the South African tunnelling species. In warmer climates, many species are active for longer periods of the year.

Dung beetles were initially introduced in the late 1960s with a view to controlling buffalo flies by removing the dung within a day or two and so preventing flies from breeding. However, other benefits have become evident. Once the beetle larvae have finished pupation, the residue is a first-rate source of fertiliser. The tunnels abandoned by the beetles provide excellent aeration and water channels for root systems. In addition, when the new generation of beetles has left the nest the abandoned burrows are an attractive habitat for soil-enriching earthworms. The digested dung in these burrows is an excellent food supply for the earthworms, which decompose it further to provide essential soil nutrients. If it were not for the dung beetle, chemical fertiliser and dung would be washed by rain into streams and rivers before it could be absorbed into the hard earth, polluting water courses and causing blooms of blue-green algae. Without the beetles to dispose of the dung, cow pats would litter pastures making grass inedible to cattle and depriving the soil of sunlight. Australia's 30 million cattle each produce 10-12 cow pats a day. This amounts to 1.7 billion tonnes a year, enough to smother about 110,000 sq km of pasture, half the area of Victoria.

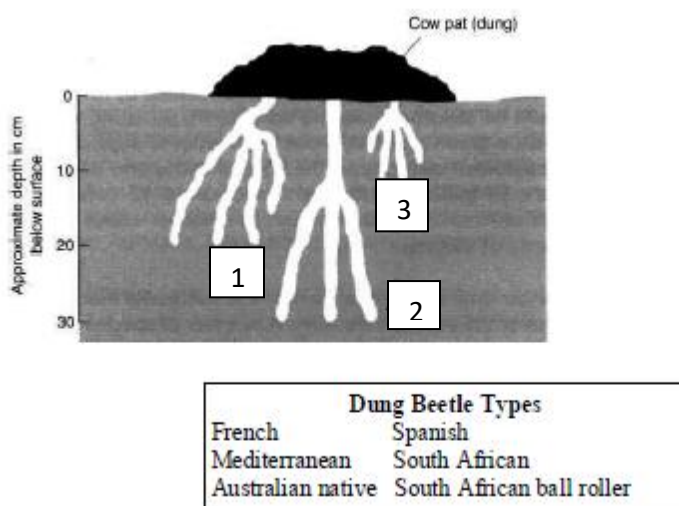
Dung beetles have become an integral part of the successful management of dairy farms in Australia over the past few decades. A number of species are available from the CSIRO or through a small number of private breeders, most of whom were entomologists with the CSIRO's dung beetle unit who have taken their specialised knowledge of the insect and opened small businesses in direct competition with their former employer.

Glossary

1. Dung: - the droppings or excreta of animals
2. Cow pats: - droppings of cows

Question 1 - 3

Label the tunnels on the diagram below. Choose your labels from the box below the diagram.



1. _____
2. _____
3. _____

Practise Reading Passage 1.3

AIRPORTS ON WATER

River deltas are difficult places for map makers. The river builds them up, the sea wears them down; their outlines are always changing. The changes in China's Pearl River delta, however, are more dramatic than these natural fluctuations. An island six kilometres long and with a total area of 1248 hectares is being created there. And the civil engineers are as interested in performance as in speed and size. This is a bit of the delta that they want to endure.

The new island of Chek Lap Kok, the site of Hong Kong's new airport, is 83% complete. The giant dumper trucks rumbling across it will have finished their job by the middle of this year and the airport itself will be built at a similarly breakneck pace.

As Chek Lap Kok rises, however, another new Asian island is sinking back into the sea. This is a 520-hectare island built in Osaka Bay, Japan, that serves as the platform for the new Kansai airport. Chek Lap Kok was built in a different way, and thus hopes to avoid the same sinking fate.

The usual way to reclaim land is to pile sand rock on to the seabed. When the seabed oozes with mud, this is rather like placing a textbook on a wet sponge: the weight squeezes the water out, causing both water and sponge to settle lower. The settlement is rarely even: different parts sink at different rates. So buildings, pipes, roads and so on tend to buckle and crack. You can engineer around these problems, or you can engineer them out. Kansai took the first approach; Chek Lap Kok is taking the second.

The differences are both political and geological. Kansai was supposed to be built just one kilometer offshore, where the seabed is quite solid. Fishermen protested, and the site was shifted a further five kilometers. That put it in deeper water (around 20 meters) and above a seabed that consisted of 20 meters of soft alluvial silt and mud deposits. Worse, below it was a not-very- firm glacial deposit hundreds of meters thick.

The Kansai builders recognized that settlement was inevitable. Sand was driven into the seabed to strengthen it before the landfill was piled on top, in an attempt to slow the process; but this has not been as effective as had been hoped. To cope with settlement, Kansai's giant terminal is supported on 900 pillars. Each of them can be individually jacked up, allowing wedges to be added underneath. That is meant to keep the building level. But it could be a tricky task.

Conditions are different at Chek Lap Kok. There was some land there to begin with, the original little island of Chek Lap Kok and a smaller outcrop called Lam Chau. Between them, these two outcrops of hard, weathered granite make up a quarter of the new island's surface area. Unfortunately, between the islands there was a layer of soft mud, 27 meters thick in places.

According to Frans Uiterwijk, a Dutchman who is the project's reclamation director, it would have been possible to leave this mud below the reclaimed land, and to deal with the resulting settlement by the Kansai method. But the consortium that won the contract for the island opted for a more aggressive approach. It assembled the world's largest fleet of dredgers, which sucked up 150m cubic meters of clay and mud and dumped it in deeper waters. At the same time, sand was dredged from the waters and piled on top of the layer of stiff clay that the massive dredging had laid bare.

Nor was the sand the only thing used. The original granite island which had hills up to 120 meters high was drilled and blasted into boulders no bigger than two meters in diameter. This provided 70m cubic meters of granite to add to the island's foundations. Because the heap of boulders does not fill the space perfectly, this represents the equivalent of 105m cubic meters of landfill. Most of the rock will become the foundations for the airport's runways and its taxiways. The sand dredged from the waters will also be used to provide a two-meter capping layer over the granite platform. This makes it easier for utilities to dig trenches –granite is unyielding stuff. Most of the terminal buildings will be placed above the site of the existing island. Only a limited amount of pile-driving is needed to support building foundations above softer areas.

The completed island will be six to seven meters above sea level. In all, 350m cubic meters of material will have been moved. And much of it, like the overloads, has to be moved several times before reaching its final resting place. For example, there has to be a motorway capable of carrying 150-tonne dump-trucks; and there has to be a raised area for the 15,000 construction workers. These are temporary; they will be removed when the airport is finished.

The airport, though, is here to stay. To protect it, the new coastline is being bolstered with a formidable twelve kilometers of sea defenses. The brunt of a typhoon will be deflected by the neighboring island of Lantau; the sea walls should guard against the rest. Gentler but more persistent bad weather - the downpours of the summer monsoon – is also being taken into account. A mat-like material called geo textile is being laid across the island to separate the rock and sand layers. That will stop sand particles from being washed into the rock voids, and so causing further settlement. This island is being built never to be sunk.

Questions 1-4

Complete the labels on Diagram B below. Choose your answers from the box below the diagram and write your answers.

- A. Granite
- B. runways and taxiways
- C. mud water
- D. terminal building site
- E. stiff clay
- F. sand

1. _____
2. _____
3. _____
4. _____

DIAGRAM A

Cross-section of the original area around Chek Lap Kok before work began

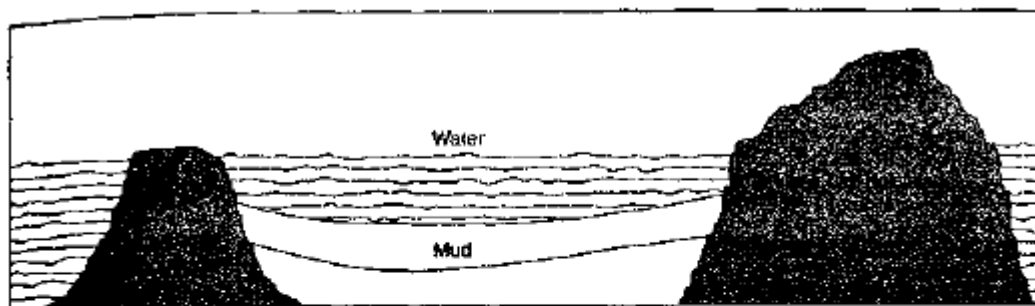
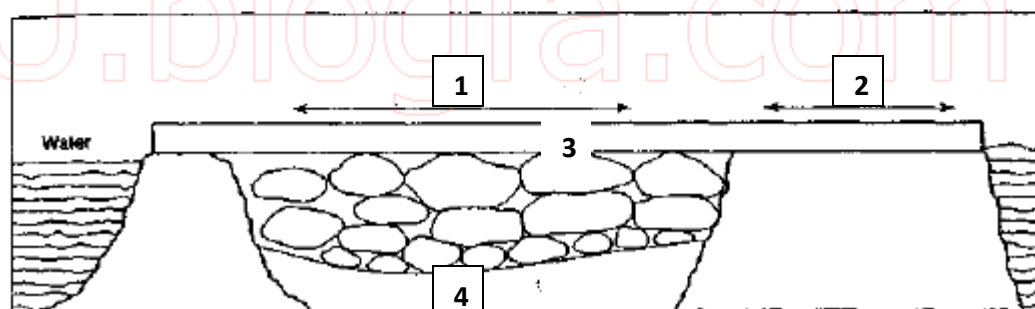


DIAGRAM B

Cross-section of the same area at the time the article was written



Practise Reading Passage 1.4

TIDAL POWER

Undersea turbines which produce electricity from the tides are set to become an important source of renewable energy for Britain. It is still too early to predict the extent of the impact they may have, but all the signs are that they will play a significant role in the future.

A. Operating on the same principle as wind turbines, the power in sea turbines comes from tidal currents which turn blades similar to ships' propellers, but, unlike the wind, the tides are predictable and the power input is constant. The technology raises the prospect of Britain becoming self-sufficient in renewable energy and drastically reducing its carbon dioxide emissions, if tide, wind and wave power are all developed. Britain would be able to close gas, coal and nuclear power plants and export renewable power to other parts of Europe. Unlike wind power which Britain originally developed and then abandoned for 20 years allowing the Dutch to make it a major industry, undersea turbines could become a big export earner to island nations such as Japan and New Zealand.

B. Tidal sites have already been identified that will produce one-sixth or more of the UK's power - and at prices competitive with modern gas turbines and undercutting those of the already ailing nuclear industry. One site alone, the Pender Firth, between Orkney and mainland Scotland, could produce 10% of the country's electricity with banks of turbines under the sea, and another at Alderney in the Channel islands three times the 1,200 megawatts of Britain's largest and newest nuclear plant, Sizewell B, in Suffolk. Other sites identified include the Bristol Channel and the west coast of Scotland, particularly the channel between Campbeltown and Northern Ireland.

C. Work on designs for the new turbine blades and sites are well advanced at the University of Southampton's sustainable energy research group. The first station is expected to be installed off Lynmouth in Devon shortly to test the technology in a venture jointly funded by the department of Trade and Industry and the European Union. AbuBakrBahaj, in charge of the Southampton research, said: 'The prospects for energy from tidal currents are far better than from wind because the flows of water are predictable and constant. The technology for dealing with the hostile saline environment under the sea has been developed in the North Sea oil industry and much is already known about turbine blade design, because of wind power and ship propellers. There are a few technical difficulties, but I believe in the next nine to ten years we will be installing commercial marine turbine farms.' Southampton has been awarded £2.15m over three years to develop the turbines and is working with Marine Current Turbines, a subsidiary of IT power, on the Lynmouth project. EU research has now identified 1GB potential sites for tidal power round the coasts of Britain. The best sites are between islands or around heavily indented coasts where there are strong tidal currents.

D. A marine turbine blade needs to be only one-third of the size of a wind generator to produce three times as much power. The blades will be about 20 metres in diameter so around 30 metres of water is required. Unlike wind power, there are unlikely to be environmental objections. Fish and

other creatures are thought unlikely to be at risk from the relatively slow turning blades. Each turbine will be mounted on a tower which will connect to the national power supply grid via underwater cables. The towers will stick out of the water and be lit. to warn shipping, and also be designed to be lifted out of the water for maintenance and to clean seaweed from the blades.

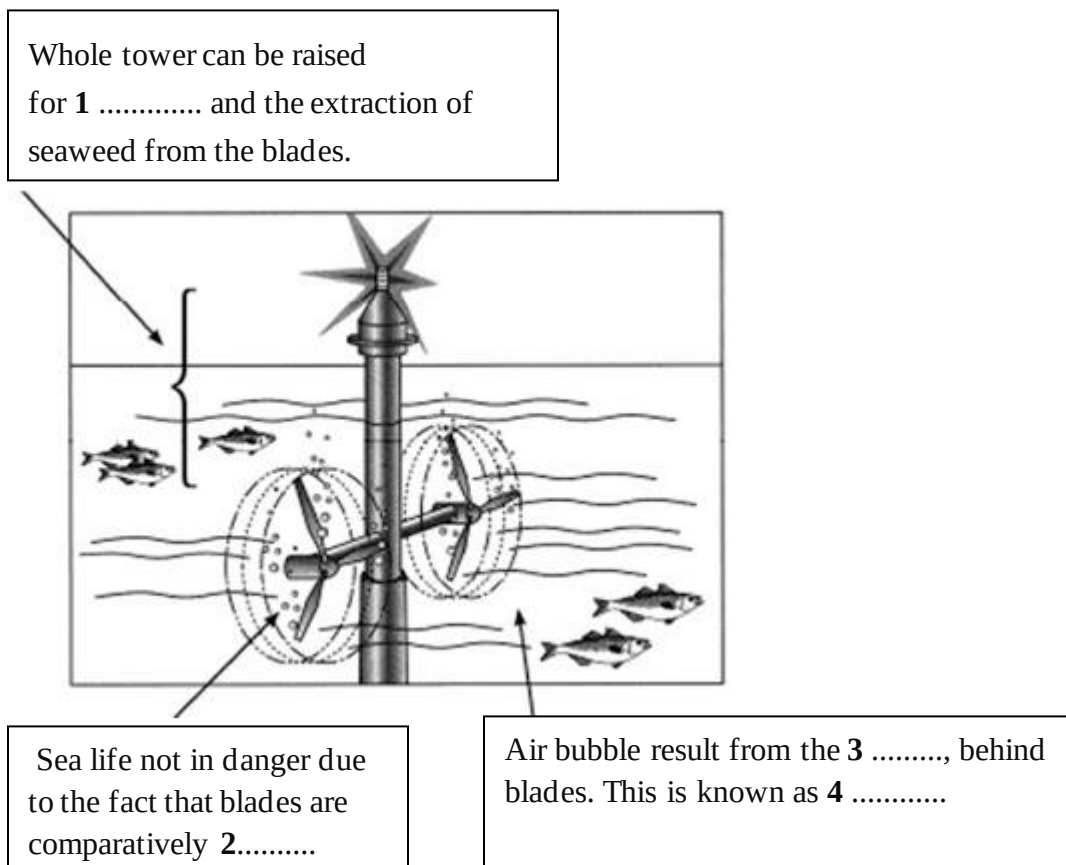
E. Dr Baha has done most work on the Alderney site, where there are powerful currents. The single undersea turbine farm would produce far more power than needed for the Channel Islands and most would be fed into the French Grid and be re-imported into Britain via the cable under the Channel.

F. One technical difficulty is cavitations, where low pressure behind a turning blade causes air bubbles. These can cause vibration and damage the blades of the turbines. Dr Bahaj said: 'We have to test a number of blade types to avoid this happening or at least make sure it does not damage the turbines or reduce performance. Another slight concern is submerged debris floating into the blades. So far we do not know how much of a problem it might be. We will have to make the turbines robust because the sea is a hostile environment. but all the signs that we can do it are good.

Questions 1 – 4

Label the diagram below: Choose **NO MORE THAN TWO WORDS** from the passage for each answer: Write your answers in boxes 1- 4 on your answer sheet,

An Undersea Turbine



Practise Reading Passage 1.5

A CHRONICLE OF TIMEKEEPING

Our conception of time depends on the way we measure it



A According to archaeological evidence, at least 5, 000 years ago, and long before the advent of the Roman Empire, the Babylonians began to measure time, introducing calendars to coordinate communal activities, to plan the shipment of goods and, in particular, to regulate planting and harvesting. They based their calendars on three natural cycles: the solar day, marked by the successive periods of light and darkness as the earth rotates on its axis; the lunar month, following the phases of the moon as it orbits the earth; and the solar year, defined by the changing seasons that accompany our planet's revolution around the sun.

B Before the invention of artificial light, the moon had greater social impact. And, for those living near the equator, in particular, its waxing and waning were more conspicuous than the passing of the seasons. Hence, the calendars that were developed at the lower latitudes were influenced more by the lunar cycle than by the solar year. In more northern climes, however, where seasonal agriculture was practised, the solar year became more crucial. As the Roman Empire expanded northward, it organised its activity chart for the most part around the solar year.

C Centuries before the Roman Empire, the Egyptians had formulated a municipal calendar having 12 months of 30 days, with five days added to approximate the solar year. Each period of ten days was marked by the appearance of special groups of stars called decans. At the rise of the star Sirius just before sunrise, which occurred around the all-important annual flooding of the Nile, 12 decans could be seen spanning the heavens. The cosmic significance the Egyptians placed in the 12 decans led them to develop a system in which each interval of darkness (and later, each interval of daylight) was divided into a dozen equal parts. These periods became known as temporal hours because their duration varied according to the changing length of days and nights with the passing of the seasons. Summer hours were long, winter ones short; only at the spring and autumn equinoxes were the hours of daylight and darkness equal. Temporal hours, which were first adopted by the Greeks and then the Romans, who disseminated them through Europe, remained in use for more than 2, 500 years.

D In order to track temporal hours during the day, inventors created sundials, which indicate time by the length or direction of the sun's shadow. The sundial's counterpart, the water clock, was designed to measure temporal hours at night. One of the first water clocks was a basin with a

small hole near the bottom through which the water dripped out. The falling water level denoted the passing hour as it dipped below hour lines inscribed on the inner surface. Although these devices performed satisfactorily around the Mediterranean, they could not always be depended on in the cloudy and often freezing weather of northern Europe.

E The advent of the mechanical clock meant that although it could be adjusted to maintain temporal hours, it was naturally suited to keeping equal ones. With these, however, arose the question of when to begin counting, and so, in the early 14th century, a number of systems evolved. The schemes that divided the day into 24 equal parts varied according to the start of the count: Italian hours began at sunset, Babylonian hours at sunrise, astronomical hours at midday and 'great clock' hours, used for some large public clocks in Germany, at midnight. Eventually, these were superseded by 'small clock', or French, hours, which split the day into two 12-hour periods commencing at midnight.

F The earliest recorded weight-driven mechanical clock was built in 1283 in Bedfordshire in England. The revolutionary aspect of this new timekeeper was neither the descending weight that provided its motive force nor the gear wheels (which had been around for at least 1,300 years) that transferred the power; it was the part called the escapement. In the early 1400s came the invention of the coiled spring or fusee which maintained a constant force to the gear wheels of the timekeeper despite the changing tension of its mainspring. By the 16th century, a pendulum clock had been devised, but the pendulum swung in a large arc and thus was not very efficient.

G To address this, a variation on the original escapement was invented in 1670, in England. It was called the anchor escapement, which was a lever-based device shaped like a ship's anchor. The motion of a pendulum rocks this device so that it catches and then releases each tooth of the escape wheel, in turn allowing it to turn a precise amount. Unlike the original form used in early pendulum clocks, the anchor escapement permitted the pendulum to travel in a very small arc. Moreover, this invention allowed the use of a long pendulum which could beat once a second and thus led to the development of a new floor-standing case design, which became known as the grandfather clock.

H Today, highly accurate timekeeping instruments set the beat for most electronic devices. Nearly all computers contain a quartz-crystal clock to regulate their operation. Moreover, not only do time signals beamed down from Global Positioning System satellites calibrate the functions of precision navigation equipment, they do so as well for mobile phones, instant stock-trading systems and nationwide power-distribution grids. So integral have these time-based technologies become to day-to-day existence that our dependency on them is recognised only when they fail to work.

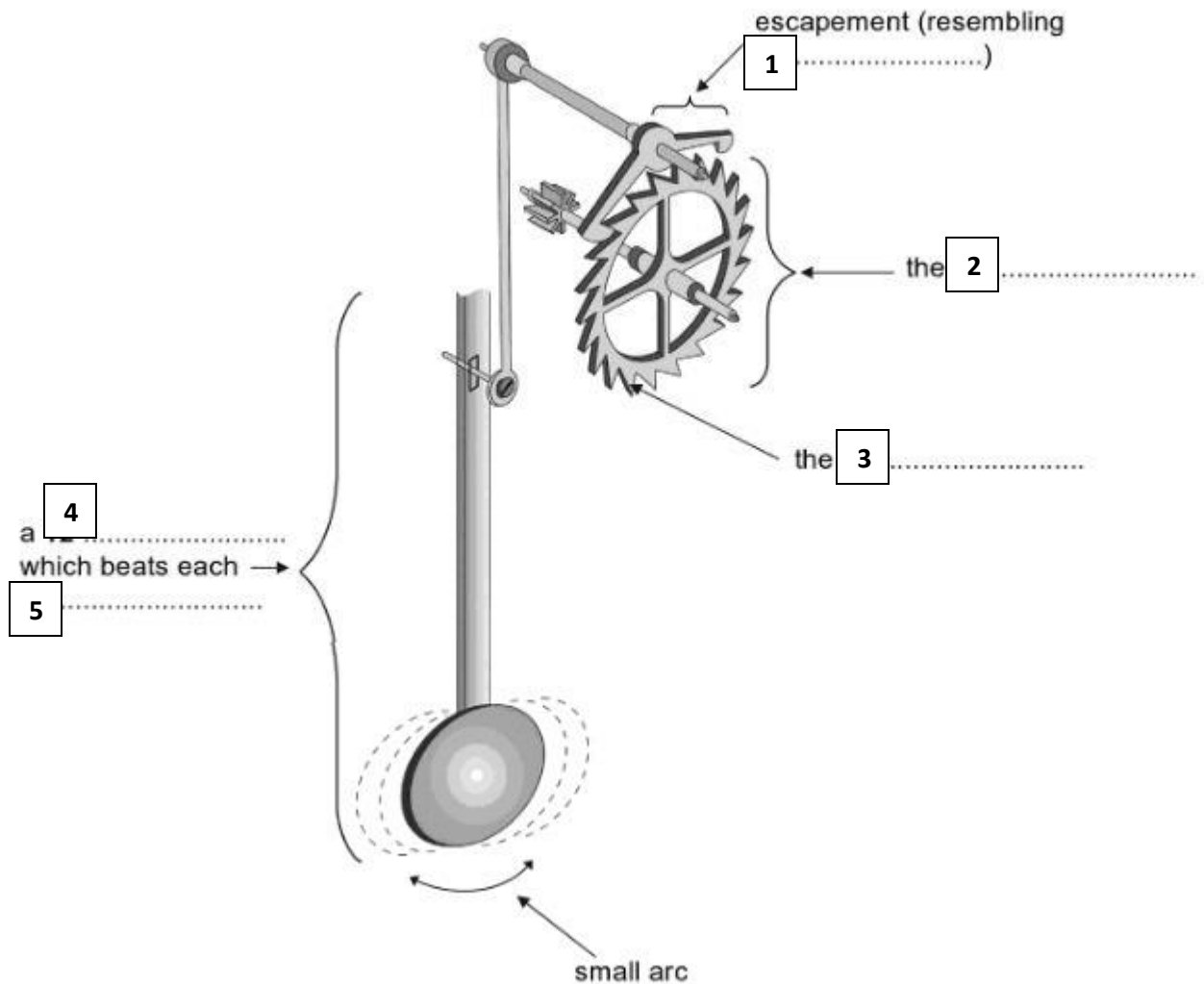
Questions 1-5

Label the diagram below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes 1-5 in your answer sheet

How the 1670 lever-based device worked



ANSWER KEY

FLOW-CHART COMPLETION TASK

PASSAGE 1.1 - EXPANDS - COLD AIR JETS - TENSILE FORCE - COMPRESSIVE FORCE - RESISTANT	PASSAGE 1.2 - TRANSMITTED - FILM // NEGATIVES - PRINTING PLATES - PROGRAMMED - WRAPPING - WEIGHED - PASTER ROBOT - STORAGE AREA	PASSAGE 1.3 - MEASUREMENTS - GRAPH - CLUSTER - AMPHIBIOUS - HALF WAY - DRY-LAND TORTOISES
PASSAGE 1.4 - INITIAL APPLICATION - WALK-IN DAY - SWIMMING TEST - VERBAL REFERENCES - RECRUITEMENT POOL - FULL INTERVIEW - EMERGENCY	PASSAGE 1.5 - PREPARE - FULL - SMOKE - CHARGE - MACHINES - COMBS - SPLIT	

DIAGRAM COMPLETION TASK

PASSAGE 1.1 1. E – USE OF BRAKES 2. C – RAPID SPINNING 3. A – STEADY SPINNING	PASSAGE 1.2 1. SOUTH AFRICAN 2. FRENCH 3. SPANISH
PASSAGE 1.3 1. RUNWAYS AND TAXIWAYS 2. TERMINAL BUILDING SITE 3. SAND STIFF CLAY	PASSAGE 1.4 1. MAINTENANCE 2. SLOW 3. LOW PRESSURE 4. CAVITATION
PASSAGE 1.5 1. ANCHOR 2. ESCAPE WHEEL 3. TOOTH 4. PENDULUM 5. SECOND	